Interaction-Aware Trajectory Prediction for Safe Motion Planning in Autonomous Driving: A Transformer-Transfer Learning Approach

Jinhao Liang, *Member, IEEE*, Chaopeng Tan, Longhao Yan, *Member, IEEE*, Jingyuan Zhou, *Graduate Student Member, IEEE*, Guodong Yin, *Senior Member, IEEE*, and Kaidi Yang, *Member, IEEE*

***Abstract*— A critical aspect of safe and efficient motion planning for autonomous vehicles (AVs) is to handle the complex and uncertain behavior of surrounding human-driven vehicles (HDVs). Despite intensive research on driver behavior prediction, existing approaches often overlook the interactions between AVs and HDVs, assuming that HDV trajectories are not influenced by AV actions. To address this gap, we present a transformer-transfer learning-based interaction-aware trajectory predictor for safe motion planning in autonomous driving, focusing on a vehicle-to-vehicle (V2V) interaction scenario involving an AV and an HDV. Specifically, we construct a transformer-based interaction-aware trajectory predictor using widely available datasets of HDV trajectory data and further transfer the learned predictor using a small set of AV-HDV interaction data. Then, to better incorporate the proposed trajectory predictor into the motion planning module of AVs, we introduce an uncertainty quantification method to characterize the predictor's errors, which are integrated into the path-planning process. Our experimental results demonstrate the value of explicitly considering interactions and handling uncertainties.**

***Index Terms*— Autonomous vehicles, interaction-aware trajectory prediction, transfer learning, uncertain quantification, motion planning.**

I. Introduction

AUTONOMOUS driving technology is pivotal in revolutionizing transportation safety and efficiency. It holds great promise for reducing accidents and congestion, as well as improving accessibility for all travelers [1], [2], [3], [4]. However, as the penetration rates of autonomous vehicles (AVs) can only increase incrementally due to the gradual enhancement of technological maturity and social acceptance, AVs are envisioned to operate in mixed traffic environments alongside human-driven vehicles (HDVs) [5], [6], [7]. One challenge associated with mixed-traffic environments is the motion planning of AVs, which involves the planning of AV trajectories considering the interactions with surrounding HDVs [8]. Since human driver behavior can be highly complex and stochastic, the key to addressing this challenge is (i) to predict HDV behavior accurately and (ii) to optimize AV trajectories based on the noisy predictions of HDV's future trajectories.

Driver behavior prediction is crucial for the deployment of AVs in mixed-traffic environments, as it enables AVs to anticipate and react to the actions of HDVs [9]. With the continuous advancement of sensor technologies and machine learning algorithms, researchers have made significant progress in modeling and predicting diverse driver behaviors, such as lane changing, braking, and turning [10]. These prediction models often leverage a combination of real-time sensor data, history driving data, and contextual information to anticipate driver intentions and actions [11], [12], [13]. Early methods utilize physics-based models [14] or conventional shallow learning algorithms [15], [16] such as Support Vector Machines, Dynamic Bayesian Networks, and Hidden Markov Models. However, these models are not able to account for the variability and unpredictability of human driving behavior. In recent years, advancements in deep learning [17], [18], [19], [20], [21] have introduced more intricate models to capture complex driver behaviors in realistic driving conditions.

Despite intensive research on driver behavior prediction, existing approaches typically ignore the *interactions* between AVs and HDVs and rely on simplified assumptions that the predicted HDV trajectories are independent of the future actions of AVs [22], [23]. In other words, existing approaches lack the capacity to incorporate "what-if" analyses, assuming fixed reactions from surrounding vehicles regardless of the ego AV's actions. Nevertheless, interactions among vehicles frequently shape one another's behaviors. Thus, *interaction-aware* trajectory prediction is crucial for the safe and efficient navigation of AVs in mixed-traffic environments. Interactions between vehicles are traditionally captured using game-theoretic approaches [24], [25], [26]. Although these methods are highly interpretable and theoretically sound, they

Received 2 November 2024; revised 1 May 2025 and 25 June 2025; accepted 28 June 2025. Date of publication 21 July 2025; date of current version 21 October 2025. This work was supported by Singapore Ministry of Education (MOE) through Academic Research Fund Tier 1 under Grant A-8000404-01-00. The Associate Editor for this article was A. Petrillo. *(Corresponding author: Kaidi Yang.)*

Jinhao Liang, Longhao Yan, Jingyuan Zhou, and Kaidi Yang are with the Department of Civil and Environmental Engineering, National University of Singapore, Singapore 119077 (e-mail: jh.liang@nus.edu.sg; longhao.yan@u.nus.edu; jingyuanzhou@u.nus.edu; kaidi.yang@nus.edu.sg).

Chaopeng Tan is with the Chair of Traffic Process Automation, Technische Universität Dresden, 01069 Dresden, Germany (e-mail: tantantan951122@gmail.com).

Guodong Yin is with the School of Mechanical Engineering, Southeast University, Nanjing 211189, China (e-mail: ygd@seu.edu.cn).

Digital Object Identifier 10.1109/TITS.2025.3588228

1558-0016 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

often suffer from high computational complexity and rely on predefined utility functions [27], [28]. A small portion of existing works have investigated interaction-aware trajectory prediction by developing deep learning-based algorithms. For example, reference [29] employs a social pooling structure to understand the spatial interactions among pedestrians, utilizing hidden states produced by an LSTM network, which is further refined by reference [30] via convolutional layers. Recent studies [31], [32], [33] present a novel Planning-Informed Trajectory Prediction (PiP) framework, which incorporates the ego vehicle's future planning information into the prediction process. PiP framework has been demonstrated to improve the accuracy of the trajectory predictor in driving scenarios involving complex interactions between vehicles. To enhance the interaction capability between vehicles and between vehicles and the road, many state-of-the-art prediction methods utilize a transformer framework [34], [35] to achieve high prediction accuracy.

However, to the best of our knowledge, existing interaction-aware deep learning-based trajectory prediction algorithms suffer from two limitations. First, existing methods [34], [35], [36], [37], [38], [39] train the trajectory predictor using only HDVs' trajectory data. This is due to the lack of interactive trajectory data between AVs and HDVs in existing trajectory databases, such as NGSIM, HighD, and Waymo. Nevertheless, studies [40] have shown that human drivers exhibit noticeably different car-following behavior when interacting with AVs compared to HDVs. Specifically, when following AVs, HDVs tend to maintain more stable and consistent headways, reflecting increased predictability and perceived safety [41], [42]. Moreover, [43], [44] also indicate that human drivers tend to maintain a longer time to collision when following AVs. This indicates a level of caution, suggesting that human drivers still exhibit mistrust or uncertainty toward AVs.

Second, recent interaction-aware algorithms [31], [32], [33] do not incorporate the uncertainty of prediction results into path planning. This is, however, important due to the large variability of HDV driving behavior. Recently, safe motion planning methods [6], [45], [46] for autonomous driving systems have been studied to enable collision avoidance and efficient navigation in dynamic environments. Reference [47] proposes a risk-sensitive reinforcement learning algorithm that integrates Lyapunov functions into SAC for safe motion planning, improving stability and efficiency of AVs. However, the safe planning of autonomous driving relies on information from surrounding vehicles, and the uncertainty in trajectory prediction can degrade algorithm performance, potentially leading to collisions.

Statement of Contribution. To address these two limitations, we propose a transfer-learning-based interaction-aware trajectory prediction and planning framework that explicitly incorporates uncertainty quantification. The contributions are two-fold. First, unlike existing trajectory predictors [34], [35], [36], [37], [38], [39] that are trained using only HDV trajectory data, we propose a novel transformer-transfer learning-based trajectory predictor that leverages a small amount of interaction data between AVs and HDVs from real-world scenarios to further enhance the adaptability of the predictor to mixed-traffic scenarios. Specifically, we construct a transformer-based interaction-aware trajectory predictor using a large dataset of HDVs' trajectory data and further transfer the learned predictor using a small set of AV-HDV interaction data to account for mixed-traffic scenarios. Second, we introduce an uncertainty quantification method to characterize the predictor's errors and integrate it into path planning. By ensuring safety constraints, the trajectory selector would balance various performance indicators to generate a reference trajectory for the tracking controller.

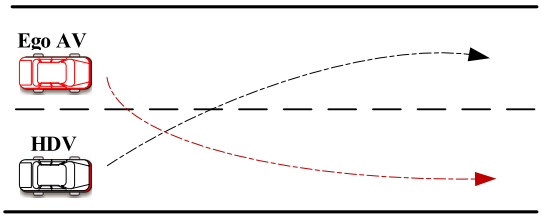


Fig. 1. V2V encounter scenario.

The remainder of this paper is organized as follows. Section II presents the methodological framework. Sections III, and IV present the designs of the trajectory predictor, path-planning algorithm, and trajectory-tracking controller, respectively. Section V performs case studies and analyzes simulation results, and Section VI concludes the paper.

II. Problem Description and Systematic Framework

A. Problem Description

As illustrated in Fig. 1, we consider a vehicle-to-vehicle (V2V) interaction scenario where an AV and an HDV seek to perform lane changes. Such a scenario is common in the early deployment stages of AVs with limited penetration rates. We treat the AV as the ego vehicle and the HDV as the encountering vehicle. The goal is to safely and efficiently control the AV, considering the potential interactions with the HDV. We make the following remarks for the studied scenario. First, we assume that the information (e.g., positions and speeds) of surrounding vehicles can be precisely measured, which is realistic thanks to the rapid advancements in autonomous driving perception systems and communication technologies [48]. Second, we assume human drivers can clearly identify that they are interacting with AVs. This assumption is realistic because (i) AVs are equipped with visible perception devices such as LiDAR, (ii) AVs being tested typically bear clear company logos, and (iii) more mature AVs may not have drivers onboard.

The dynamics of the AV are characterized by a widely used bicycle model [49] demonstrated in Figure 2 with model parameters provided in Table I. Specifically, the vehicle dynamics can be represented by a state-space equation as follows.

$$\dot{\xi} = A\xi + Bu, \tag{1}$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

TABLE I
VEHICLE CONFIGURATION PARAMETERS

Symbol	Description	Value[units]
J_z	Inertia yaw moment of the vehicle	606.1 $(\mathrm{kg}\cdot\mathrm{m}^2)$
m	Vehicle mass	1274 (kg)
c_f	Cornering stiffness of front tire	85000 (N/rad)
c_r	Cornering stiffness of rear tire	112000(N/rad)
l_f	Distance from CoG to front axle	1.016 (m)
l_r	Distance from CoG to rear axle	1.562 (m)

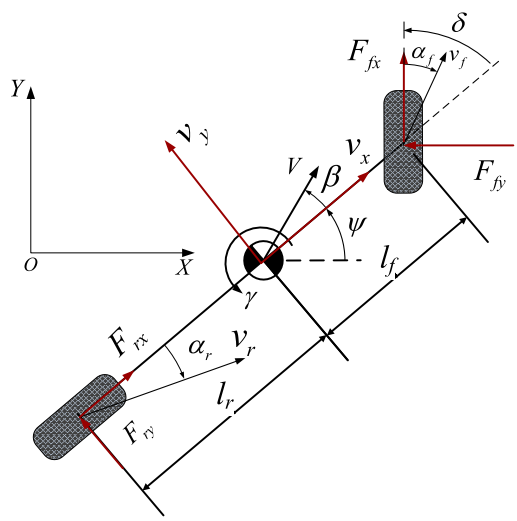


Fig. 2. Vehicle dynamics model.

with

$$A=\begin{bmatrix} 0 & 0 & 1 & 0 & -v_y & 0 \\ 0 & 0 & 0 & 1 & v_x & 0 \\ 0 & 0 & 0 & 0 & 0 & v_y \\ 0 & 0 & 0 & \frac{2(c_r+c_f)}{mv_x} & 0 & \frac{2(c_f c_f - c_r l_r)}{mv_x} - v_x \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \frac{2(c_f l_f - c_r l_r)}{J_z v_x} & 0 & \frac{2\left(c_r l_r^2 + c_f l_f^2\right)}{J_z v_x} \end{bmatrix},$$

$$B=\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{-2c_f}{m} & 0 & \frac{-2c_f l_f}{J_z} \end{bmatrix}^T,$$

where the vehicle state $\xi = [x, y, v_x, v_y, \psi, \gamma]^T$ includes the AV's global longitudinal position x, global lateral position y, longitudinal velocity v_x, lateral velocity v_y, yaw angle ψ, and yaw rate γ. The control input of the AV is represented by $u = [a_x, \delta]^T$, where a_x and δ indicate the longitudinal acceleration and steering input, respectively. Variables m and J_z are the vehicle mass and the inertia moment of yaw direction, respectively, c_f and c_r denote the cornering stiffness of the front and rear tires, respectively, and l_f and l_r indicate the distances from the vehicle center of gravity (CoG) to the front and rear axles, respectively.

B. Methodological Framework

Our methodological framework is illustrated in Fig. 3, which involves an HDV prediction module (Section III) and an AV path-planning module (Section IV).

The HDV prediction module aims to predict the future trajectories of the HDV, which serves as a crucial input to the AV motion planning module. Here, the main challenge is that the behavior of the HDV highly depends on the decisions made by the AV, which requires the trajectory prediction algorithm to be interaction-aware. Moreover, the HDV response to the AV may be different from its response to another HDV. Therefore, the HDV prediction model trained fully based on HDV naturalistic driving data, as constructed in most existing studies, may not be suitable to characterize the HDV's interactions with the AV. To address these challenges, we first train an interaction-aware transformer-based predictor based on a real-world HDV driving dataset and then leverage a transfer learning framework to derive the HDV model that better characterizes HDV's interactions with AVs, using limited autonomous driving data.

The AV path-planning module would generate the candidate paths and then select the optimal reference trajectory from these paths. Since the AV path-planning module takes the HDV trajectory prediction as input, its performance significantly relies on the accuracy of the prediction. To address this challenge, we explicitly quantify the uncertainties of the prediction module by evaluating the prediction module's performance on online data. Such uncertainties are explicitly incorporated into the motion planning of AVs to derive the reference trajectory, which is tracked via a model predictive controller (MPC).

III. A TRANSFORMER-TRANSFER LEARNING-BASED TRAJECTORY PREDICTOR FOR HDVS

This section introduces an interaction-aware HDV trajectory prediction model that considers the potential response of HDV trajectories to AV decisions. Section III-A presents a transformer-based interaction-aware HDV trajectory prediction model, and Section III-B introduces a transfer learning approach for training the proposed trajectory prediction model to adapt to the limited availability of the interaction data between HDVs and AVs. Section III-C presents a quantification method to describe the uncertain prediction results.

A. Interaction-Aware Trajectory Prediction Model

In this section, we propose an interaction-aware transformer-based model for predicting the future trajectories of HDVs, which captures the response of HDVs to AV decisions by including the planned trajectories of AVs as input to the HDV trajectory prediction model. Mathematically, let X^t_{AV} and X^t_{HDV} represent the trajectory information of the AV and the HDV at time step t, respectively. Specifically, the trajectory information includes both the longitudinal and lateral positions, as well as the corresponding longitudinal and lateral velocities. Then, we use the following information to predict the HDV's future trajectory $\hat{X}^t_{\mathrm{HDV,P}} = \{\hat{X}^{t'}_{\mathrm{HDV}}\}^{t+t_p}_{t'=t+1}$: (i) the history trajectory of the AV, denoted by $X^t_{\mathrm{AV,H}} = \{X^{t'}_{\mathrm{AV}}\}^{t}_{t'=t-t_h+1}$, and the history trajectory of the HDV, denoted

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

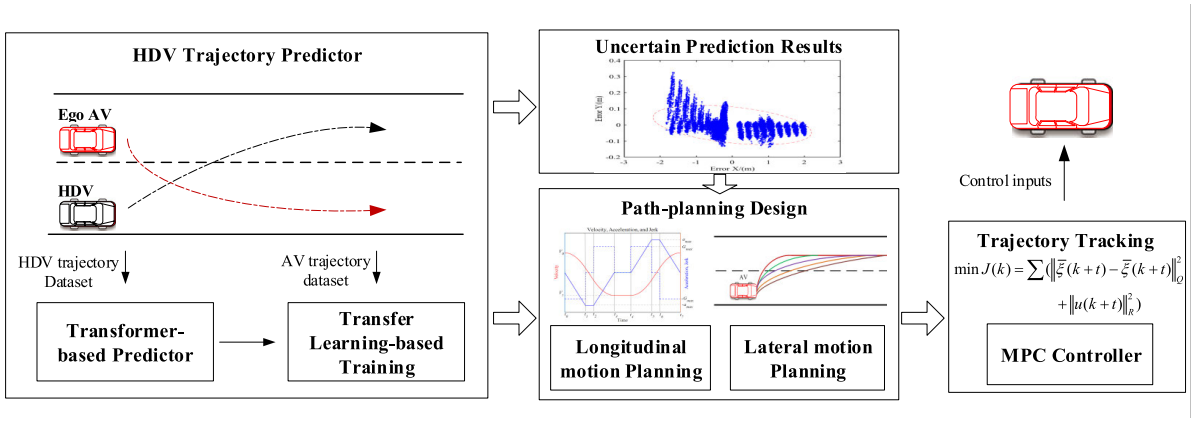


Fig. 3. The systematic framework.

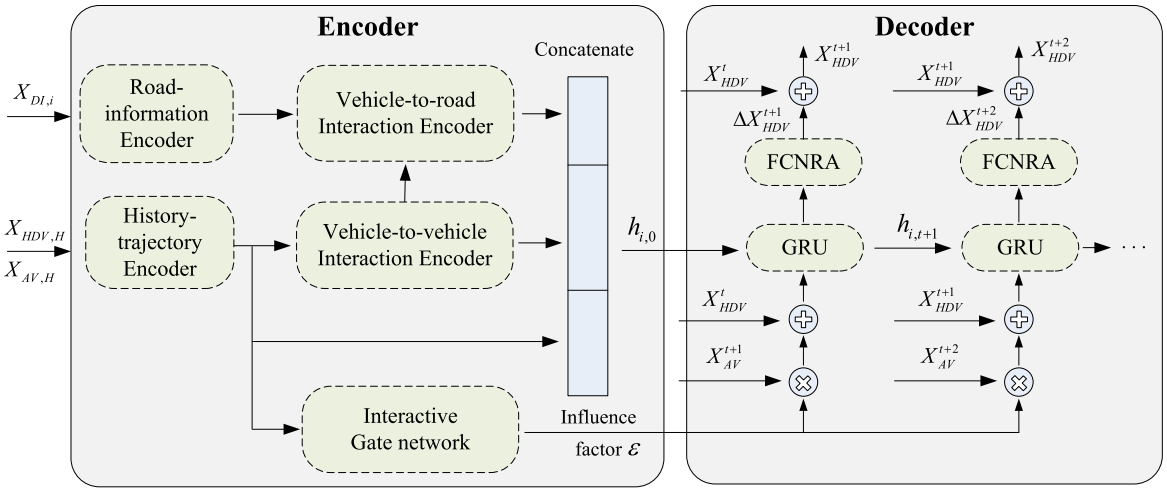


Fig. 4. Interaction-aware based predictor.

by $X^t_{\text{HDV,H}} = \{X^{t'}_{\text{HDV}}\}^t_{t'=t-t_h+1}$, (ii) road information, and (iii) the AV's planned trajectory $X^t_{\text{AV,P}} = \{X^{t'}_{\text{AV}}\}^{t+t_p}_{t'=t+1}$, where t_h and t_p represents the time horizons of history data and predictions, respectively. Notice that the notation $\hat{X}$ (with the hat symbol) represents the predicted values while the notation X (without the hat symbol) represents information known at time step t.

The detailed architecture of the proposed trajectory predictor model is illustrated in Fig. 4, which includes an encoder component and a decoder component. As shown in Fig. 4, the encoder component consists of a history trajectory encoder, a road information encoder, and an interaction encoder between vehicles and the road.

1) History Trajectory and Road Information Encoder: The history trajectory encoder uses a simple transformer-based architecture consisting of a positional encoding module, a self-attention module, and a feed-forward neural network module. First, a positional encoding module is added to the history trajectories of AVs ($X_{\text{AV,H}}$) and HDVs ($X_{\text{HDV,H}}$) to construct the input sequence. Then, the self-attention module is used to extract deep connections within the input sequence across different historical moments. Finally, we utilize a feed-forward neural network to obtain the feature representation, with the output of the last time step serving as the final feature representation to design the interaction encoder.

The road information encoder consists of two fully connected layers with ReLU activation functions to extract road features. In actual driving conditions, we are most concerned about the road information of the road center, road boundaries, and speed limit. Therefore, we represent the road information as multi-sets of points along the path, with each path point characterized by unique features $X_{\text{DI},i} = (X_{\text{cen},i}, X_{\text{left},i}, X_{\text{righgt},i}, V_{\text{lim},i})$, where $X_{\text{cen},i}$, $X_{\text{left},i}$, $X_{\text{right},i}$, and $V_{\text{lim},i}$ represent the road center position, left boundary position, right boundary position, and speed limit at the i-th sample point, respectively. The position variables include both the longitudinal and lateral coordinates.

2) Interaction Encoder: Interaction behavior is modeled using both a V2V interaction encoder and a vehicle-to-road interaction encoder. The key to modeling these interactions in trajectory prediction lies in capturing the mutual influences and dynamic dependencies among agents to accurately predict their future behaviors. The transformer's attention mechanism enhances this process by enabling the model to focus on relevant interactions, thereby improving the prediction of complex behaviors through a better understanding of how each agent influences others in dynamic environments [50]. To model

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

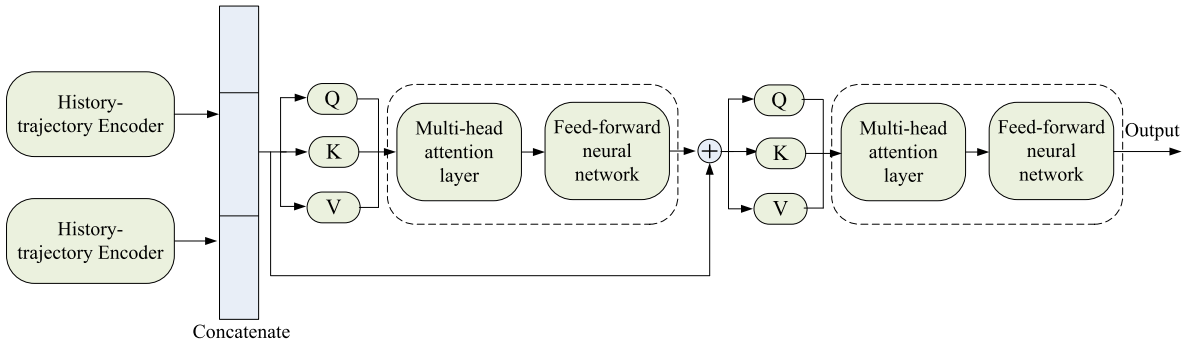


Fig. 5. Vehicle-vehicle interaction encoder.

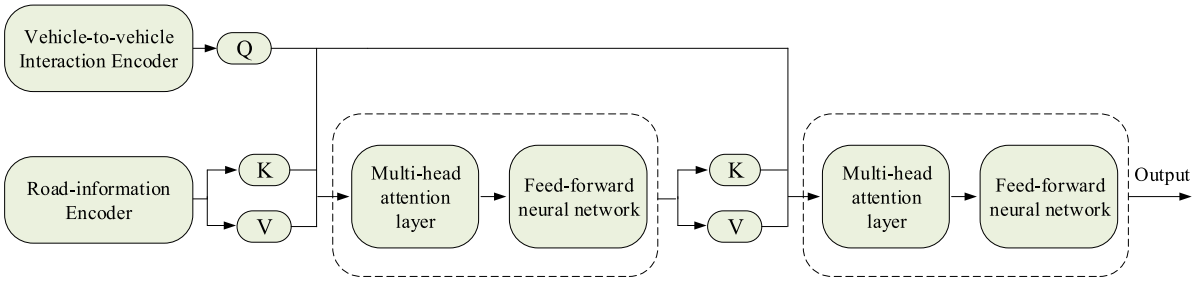


Fig. 6. Vehicle-road interaction encoder.

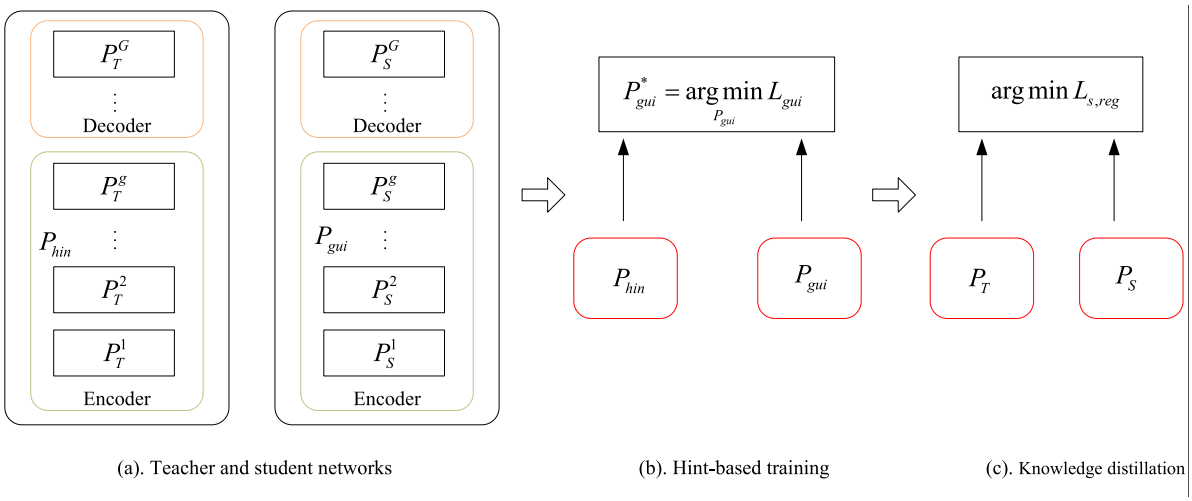


Fig. 7. The training process of the transfer-learning framework.

this, the attention mechanism computes the similarity between query (Q) and key (K) to generate attention weights, which are then used to aggregate value (V) and extract relevant information. This allows the model to effectively capture dependencies between different input elements and refine trajectory predictions [51].

Specifically, we adopt a two-layer self-attention module to extract vehicle-to-vehicle interaction features as shown in Fig. 5. Each self-attention module consists of a multi-head attention layer and a feed-forward neural network. The history trajectory information of the two interacting HDVs is concatenated, and the fused features are used simultaneously as the query, key, and value inputs for the self-attention module. The two-layer self-attention modules are connected using a residual connection architecture, which allows for the integration of the original input features into the second layer and helps the model more accurately extract and understand the relationships between vehicles in complex interaction scenarios [52].

Similar to the V2V interaction module, the vehicle-to-road interaction encoder (see Fig. 6) includes two cross-attention modules to extract interaction information. These cross-attention modules can effectively combine information from different sources (such as V2V interactions and vehicle-to-road environment interactions), capturing important relationships and thereby improving prediction accuracy. Each cross-attention module consists of a multi-head attention layer and a feed-forward neural network. The V2V encoder output, serving as a query, enables the model to focus on dynamic interactions between vehicles, while the feature vectors from the road-information encoder, used as both key and value, enhance the model's understanding of road information and environmental constraints. Then, two-layer

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

cross-attention modules are employed to extract the interaction features between driving agents and roads.

Finally, the output from the history-trajectory encoder, V2V encoder, and vehicle-to-road encoder are concatenated into h_{i0} and fed into the decoder to predict the HDVs' future trajectories.

3) Decoder Design: We next present the decoder, as shown in Fig. 4, which converts the output of the encoder to the predicted future HDV trajectories. Specifically, the main component of the decoder is t_p gated recurrent units (GRUs), each corresponding to a future time step $t' = t+1, \cdots, t+t_p$. These GRUs are chained through the hidden states, which are initialized by the concatenated output from the encoder component (denoted by h_i^0). The input to the GRU corresponding to time step t' is designed to be a weighted combination of the HDV state at time step $t'-1$ and the planned AV state at time step t', i.e.,

$$X_{\text{GRU,in}}^{t'} = \varepsilon X_{\text{AV}}^{t'} + X_{\text{HDV}}^{t'-1} \tag{2}$$

It is worth noting that the future planned trajectory of the AV is introduced into the GRU to enable interaction awareness among HDVs. The weight parameter ϵ ranging between 0 and 1 is derived from an interactive gate network in the encoder component, which builds on a fully connected network with the output mapped to the interval [0, 1] via a sigmoid function. In complex traffic environments, predicting future trajectories involves uncertainty. The introduction of the weight parameter ε allows the model to adjust its evaluation of different trajectory possibilities based on the current environment and history behavior, thereby better addressing this uncertainty [53].

The hidden state of each GRU is further processed through a single-layer fully connected network with a ReLU activation function (FCNRA) to obtain the predicted change in HDV states $\Delta X_{\text{HDV}}^{t'}$, which is further accumulated to calculate the predicted HDV state $\hat{X}_{\text{HDV,P}}^{t} = \{X_{\text{HDV}}^{t'}\}_{t'=t+1}^{t+t_p}$.

B. Transfer Learning-Based Training Approach

One challenge for training the interaction-aware trajectory prediction model presented in Section III-A is the limited availability of the interaction data between HDVs and AVs. To address this challenge, we propose a transfer learning framework that adapts the model trained on HDV trajectory data to capture the interactions between AVs and HDVs. Specifically, we first use a large amount of HDVs' trajectory data to train the prediction model, which serves as a teacher network. Then, we leverage a transfer learning framework and a small amount of interaction data between HDVs and AVs to adapt the teacher network to a student network with a similar architecture. The reason for adopting the transfer learning framework is based on the observation that the HDV-AV interactions and HDV-HDV interactions share similarities in the fundamental reasoning processes of human drivers. Consequently, the transfer learning framework can exploit knowledge learned about HDV interactions to accelerate the training of the interaction-aware HDV prediction model in mixed-traffic environments, despite limited autonomous driving data.

Due to the modular design of the trajectory predictor, which includes an encoder and a decoder, it becomes feasible to incorporate a hint-based training process [54], [55] within the traditional knowledge distillation framework [56]. In this modular transfer-learning process, we utilize not only the final output of the whole teacher network but also the intermediate representations learned by the encoder as hints to enhance the training effectiveness of each module and improve the overall performance of the predictor [55].

1) Classical Knowledge Distillation Process: Knowledge distillation is a machine learning technique where a large, complex model (the teacher) transfers its knowledge to a smaller, simpler model (the student) [56]. Hence, learning a good regression model from the teacher network is important to ensure prediction accuracy of the student network. However, the teacher's regression guidance may sometimes contradict the true reference direction [57]. Therefore, the output of the teacher's regression model is used as an upper limit for the student model to achieve, rather than as a reference value in this work. Typically, we expect the student's regression model to align closely with the true reference value. Once the performance of the student network behaves better than that of the teacher by a certain margin, we will not impose additional loss on the student model. It is called the bounded regression loss L^T of the teacher model, which serves as the foundation for constructing the regression loss L_{reg}^S of the student model.

$$L^T(\hat{X}_{\text{HDV,P}}^T, \hat{X}_{\text{HDV,P}}^S, X_{\text{HDV}}^R) = \begin{cases} ||\hat{X}_{\text{HDV,P}}^S - X_{\text{HDV}}^R||_2^2, & \text{if } ||\hat{X}_{\text{HDV,P}}^S - X_{\text{HDV}}^R||_2^2 + n > ||\hat{X}_{\text{HDV,P}}^T - X_{\text{HDV}}^R||_2^2 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

$$L_{\text{reg}}^S = L_{\text{smo}}(\hat{X}_{\text{HDV,P}}^S, X_{\text{HDV}}^R) + \nu L^T(\hat{X}_{\text{HDV,P}}^T, \hat{X}_{\text{HDV,P}}^S, X_{\text{HDV}}^R) \tag{4}$$

where $\hat{X}_{\text{HDV,P}}^T$ and $\hat{X}_{\text{HDV,P}}^S$ represent the prediction values of the teacher network and student network, respectively. X_{HDV}^R is the true reference value. L_{reg}^S is the constructed regression loss for the student network used for the training. n indicates the boundary value, and v is a scale factor (set as 0.4 in this work). From Eqs. (3)-(4), it is clear that when the prediction error of the student network exceeds that of the teacher network, the regression loss L_{reg}^S will include the loss of the teacher network. Furthermore, L_{smo} represents the smooth L_s loss. i.e., $(||\hat{X}_{\text{HDV,P}}^S - X_{\text{HDV}}^R||_2^2)$, which aims to improve the training accuracy [58], where

$$L_{smoL_s} = \begin{cases} 0.5 * ||\hat{X}_{\text{HDV,P}}^S - X_{\text{HDV}}^R||_2^2, & if |\hat{X}_{\text{HDV,P}}^S - X_{\text{HDV}}^R| < 1 \\ |\hat{X}_{\text{HDV,P}}^S - X_{\text{HDV}}^R| - 0.5, & otherwise. \end{cases} \tag{5}$$

Our integrated loss motivates the student network to approximate or exceed the teacher network's performance in regression, without excessively urging the student once parity with the teacher is achieved.

2) Hint-Based Training Process: The classical knowledge distillation process only employs the final output of the

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

network. Reference [54] illustrates that leveraging the intermediate outputs (i.e., hints) provided by the teacher network can improve the training process and enhance the performance of the student network. Hints provide intermediate features and gradient information, improving gradient flow and information transmission during training, thereby enhancing the depth and effectiveness of networks. Additionally, leveraging hints allows for more efficient use of existing data, thereby reducing the need for large-scale datasets [55]. Inspired by this, we introduce a hint-based training process to improve network performance. Specifically, we utilize the hidden layer's output of the teacher network (which is called the hint layer) to guide the training of the student network (which is called the guided layer). In this work, we choose the encoder components as the hint and the guided layers for the teacher and student networks, respectively. Then we train the network of the guided layer using the following loss function.

$$L_{gui} = \frac{1}{2}||N_{hin}(P_{hin}) - N_{gui}(P_{gui})||_2^2 \tag{6}$$

where N_{hin} and N_{gui} denote the nested functions of the hint and guided layers' networks, respectively, characterized by network parameters P_{hin} and P_{gui}. Specifically, the nested functions in this work are constructed using the concatenated features h_{i0} of the encoder.

3) The Combined Training Process: We combine the aforementioned training methods, i.e., classical knowledge distillation and hint-based training, to construct the transfer learning framework. An overview of the transfer learning-based training process is provided in Fig. 7, where P_T and P_S indicate all network parameters of the teacher and student networks, respectively. First, we randomly initialize the student network parameters and train the teacher network parameters using a large amount of HDV's trajectory data (Fig. 7 (a)). Once the teacher network is trained, we extract its encoder and utilize it for hint-based training, enabling the student network to leverage intermediate representations. During this process, we obtain the guiding layer parameters P_{gui} for the student network using limited interaction data between HDVs and AVs (Fig. 7 (b)). Finally, we employ the knowledge distillation method to train the entire network, leveraging the pre-trained guiding layer parameters to facilitate efficient knowledge transfer and improve overall performance (Fig. 7 (c)). Algorithm 1 provides a detailed outline of the training process.

Algorithm 1 The Algorithm Flow of the Transfer Learning Framework

Input: The trained network parameters P_T, and P_{hint}.
Initialize: P_S, and P_{gui}
1: $P_{hin} \leftarrow (P_{hin}^1, P_{hin}^2, \ldots, P_{hin}^g)$
2: $P_{gui} \leftarrow (P_{gui}^1, P_{gui}^2, \ldots, P_{gui}^g)$
3: $P_{gui}^* \leftarrow \underset{P_{gui}}{argmin}\, L_{gui}$
4: Update the network parameters P_{gui} for the student model as P_{gui}^*
5: Perform $\arg\min L_{s,reg}$
6: **Output:** P_S^*

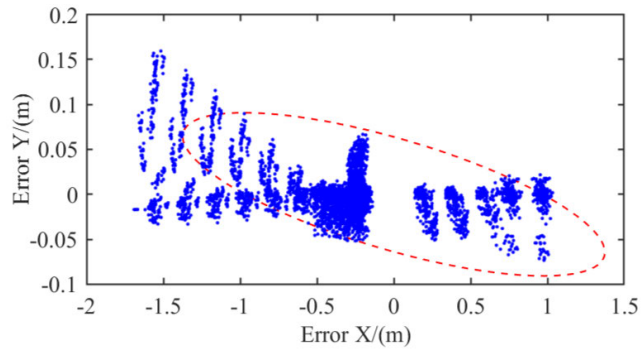


Fig. 8. The uncertain representation of the predictor on the Waymo open dataset.

C. Uncertainty Quantification of Predictor

Although the proposed trajectory predictor can improve HDV trajectory prediction accuracy in mixed traffic by explicitly accounting for the interactions between AVs and HDVs, trajectory prediction errors are still inevitable, which will further deteriorate the path planning performance of AVs and even compromise the safety of AVs. To mitigate the impact of the prediction errors on AV path planning, we explicitly quantify such errors and incorporate the uncertainty quantification results into the AV path planning described in Section IV. Specifically, this work introduces an error ellipse to describe prediction errors in both the x and y directions. Note that our approach dynamically quantifies the prediction errors based on the real-time performance of the predictors, which significantly differs from conventional methods (e.g., expanding vehicle body boundaries [59]) employing a fixed margin that does not respond to real-time conditions.

First, an error set is obtained by testing the trained network on the validation data. As the AV's longitudinal and lateral motions are highly correlated, we utilize the variance ϑ_x and ϑ_y and the covariance ϑ_{x-y} to construct a matrix Θ as follows.

$$\Theta = \begin{bmatrix} \vartheta_x^2 & \vartheta_{x-y} \\ \vartheta_{x-y} & \vartheta_y^2 \end{bmatrix},$$

Then, through the eigenvalue transformation, the error ellipse is determined as follows to quantify the uncertainty of the predictor.

$$\frac{x^2}{\eta_1} + \frac{y^2}{\eta_2} = \digamma \tag{7}$$

where η_1 and η_2 represent the largest and smallest values of the eigenvalues for the matrix Θ, respectively. $\digamma$ indicates the chi-square value, which is detailed in [60]. In this work, the chi-square value is selected to ensure a 95% confidence level, taking into account the search efficiency of the error set, whereby a higher confidence level implies a greater coverage of the error set within the error ellipse. Furthermore, the rotation angle of the ellipse can be further calculated by:

$$\psi = \arctan \frac{\tau_1}{\tau_2} \tag{8}$$

where τ_1 and τ_2 represent the basic value and the other element of the maximum eigenvector, respectively. To visually

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

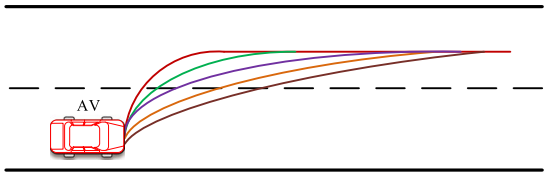


Fig. 9. The lateral motion planning.

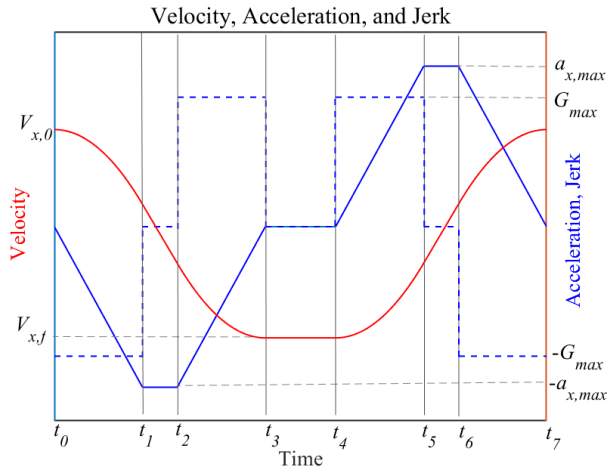


Fig. 10. The longitudinal motion planning.

present the uncertain prediction results, we utilize the Waymo Open Dataset to train the predictor and then construct the error ellipse shown in Fig. 8. Considering the uncertain prediction results, the constructed ellipse error would be utilized as a constraint to ensure the safety of AVs when generating candidate paths in Section IV-C.

IV. Path Planning and Trajectory Tracking Modules Design

We present a path-planning method for generating candidate paths in this section, which will be incorporated into the trajectory predictor in Section III. When selecting the optimal path, we incorporate the uncertainty of trajectory predictions as a safety constraint to enhance the AV's obstacle avoidance capability. After generating an optimal reference trajectory, a trajectory tracking controller is designed using the MPC-based control method.

A. Lateral and Longitudinal Trajectory Planning

The traditional trajectory generation method includes both the longitudinal and lateral motion [61] of the AV. In this work, a widely used polynomial representation [62] as shown in Fig. 9 is employed to determine the lateral position, which can be described as follows.

$$y_{\text{ref}} = \frac{10H}{Q^3}x_{\text{ref}}^3 - \frac{15H}{Q^4}x_{\text{ref}}^4 + \frac{6H}{Q^5}x_{\text{ref}}^5 \tag{9}$$

where Q and H indicate the longitudinal and lateral lane-change distances, respectively. The longitudinal lane-change distance x_{ref} depends on the AV's velocity planning. In a typical lane-change scenario, the HDVs would decelerate first to maintain lateral stability and then accelerate to maintain the initial speed control intention. Hence, a trapezoidal acceleration profile [63] is employed in this work to determine the AV's longitudinal position. Fig. 10 depicts a velocity profile that integrates acceleration and jerk. where $a_{x,max}$ and G_{max} represent the maximum values of the acceleration and jerk, respectively, while $v_{x,0}$ and $v_{x,f}$ denote the maximum and minimum velocities, respectively. The reference longitudinal position can be calculated as follows:

$$x_{ref,t} = x_{ref,t-1} + v_{x,t-1}\Lambda t + \frac{1}{2}a_{x,t-1}\Lambda_t^2 + \frac{1}{6}G_t\Lambda t^3, \tag{10}$$

where $a_{x,t}$, $v_{x,t}$, $x_{ref,t}$ and G_t represent the AV's longitudinal acceleration, longitudinal velocity, longitudinal position, and jerk at time t, respectively, and Λt indicates the time interval.

B. The Optimal Trajectory Selection

From the Waymo Open Dataset, approximately 85% of lane-change maneuvers occur within the range of 2.5 to 6.5s. Hence, we choose a constant lane-change time with an average value of 4.5s for the AV. Moreover, in lane-change scenarios, vehicle dynamics impose strong constraints on maneuver timing and controllability [64]. Accordingly, many existing studies adopt a fixed lane-change time to simplify planning and ensure safe execution [65], [66]. In this work, we adopt a similar approach by adopting a fixed lane-change time. Subsequently, candidate paths are generated by configuring different longitudinal speed profiles. The vehicle state on the candidate paths should satisfy the following constraint on vehicle dynamics [2].

$$\beta_{\text{m}} \leq \left|\tan^{-1}(0.02\mu g)\right| \tag{11}$$

where β_m indicates the AV's side slip angle at the sampling point m on the planned path, which can be calculated by $arctan\frac{v_{y,m}}{v_{x,m}}$. μ is the tire-road friction coefficient. Specifically, if the vehicle state on the reference path does not satisfy Eq. (11), it is considered to have exceeded its physical limits. Consequently, such candidate paths are physically infeasible.

Furthermore, a safe distance constraint between the AV and HDV should also be satisfied on the candidate paths. However, the HDV trajectory obtained through the trajectory predictor is not accurate. To ensure the safety of the planned path, we incorporate the error ellipse constructed in Section III-C, in addition to the safe distance threshold S_{saf}. This means that on the candidate paths, the distance S between the AV's planned trajectory and the HDV's predicted trajectory should be no lower than the sum of the axis distance L_{ell} of the error ellipse constructed in Section III-C and the safety threshold S_{saf}.

$$S \geq (L_{\text{ell}} + S_{\text{saf}}) \tag{12}$$

After generating a set of candidate paths, the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method [61] is employed to determine the optimal reference trajectory. We first need to define the following indicators to evaluate the quality of the candidate paths, including vehicle safety, stability performance, and driving comfort.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

1) Vehicle Safety: The vehicle safety is quantified by considering the potential collision with the HDVs, which is represented as follows:

$$O_{saf} = \sum_{m=1}^{\epsilon} \exp\left[\zeta_1 \left(x_{\text{ref},m} - \widetilde{x}_{\text{HDV},m}\right)^2 + \zeta_2 \left(y_{\text{ref},m} - \widetilde{y}_{\text{HDV},m}\right)^2\right] \tag{13}$$

where ϵ is the total number of sampling points. $x_{\text{ref},m}$ and $y_{\text{ref},m}$ indicate the AV's longitudinal and lateral positions at the sampling point m. Note that the HDV's longitudinal and lateral positions represented by $x_{\text{HDV},m}$ and $y_{\text{HDV},m}$ are obtained through the transformer-transfer learning-based predictor.

Considering uncertain prediction results of the HDVs' trajectory, we utilize the modified variables $\widetilde{x}_{\text{HDV},m}$ and $\widetilde{y}_{\text{HDV},m}$ to represent the HDV's longitudinal and lateral positions, respectively. where the point $(\widetilde{x}_{\text{HDV},m}, \widetilde{y}_{\text{HDV},m})$ is located on the error ellipse, which is the nearest point to the AV's reference trajectory $(x_{\text{ref},m}, y_{\text{ref},m})$. Parameters ζ_1 and ζ_2 are weight coefficients utilized to standardize the impact of longitudinal and lateral space between the AV and the HDV. Specific configurations can refer to the work [2].

2) Vehicle Stability: The vehicle stability performance is evaluated by introducing the side slip angle β [49], which is represented as follows:

$$O_{sta} = \sum_{m=1}^{\epsilon} \beta_m^2 \tag{14}$$

3) Driving Comfort: The driving comfort is quantified by the change rate of heading angle $\Delta\theta_m$ and expressed as follows.

$$O_{com} = \sum_{m=1}^{\epsilon} (\Delta\theta_m^2 + a_{x,m}^2) \tag{15}$$

After defining the performance indicators for each candidate path, the TOPSIS strategy aims to select the optimal path by selecting the one that is closest to the ideal solution [61]. Furthermore, to ensure a fair selection of the optimal path using the TOPSIS strategy, we employ the Entropy Weight Method [67], which objectively assigns weights to the three performance indicators based on their level of variability in the AV trajectory data from the Waymo Open Dataset.

C. Trajectory Tracking Controller Design

After generating an optimal reference trajectory, a trajectory tracking controller is designed using the MPC-based control method. The MPC controller aims to track the reference by constructing a quadratic programming (QP) optimization problem as follows.

$$\min\ J(k) = \sum_{t=1}^{N_p} \left(\left\|\overline{\xi}(k+t \mid k) - \overline{\xi}_{ref}(k+t \mid k)\right\|_Q^2 + \left\|u_(k+t \mid k)\right\|_R^2 \right) \tag{16a}$$

$$\text{s.t. } \mathrm{a_{x,min}} \le \mathrm{a_x(k)} \le \mathrm{a_{x,max}} \tag{16b}$$

$$\delta_{\min} \le \delta(k) \le \delta_{\max} \tag{16c}$$

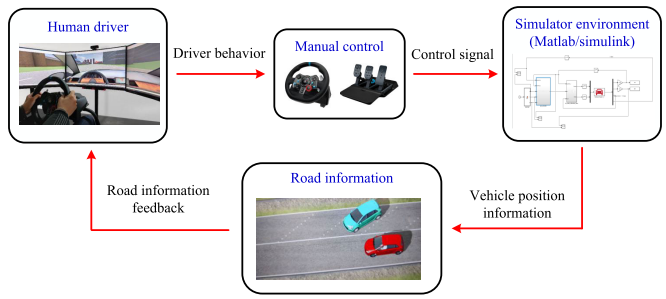


Fig. 11. The simulator setting.

$$|\gamma| \le \frac{\mu g}{V_x} \tag{16d}$$

$$\left|\alpha_f\right| \le \arctan \frac{\mu m g l_r}{2c_f\left(l_f + l_r\right)} \tag{16e}$$

$$|\alpha_r| \le \arctan \frac{\mu m g l_f}{2c_r\left(l_f + l_r\right)} \tag{16f}$$

where $\overline{\xi} = [v_x, x, y]^T$, and $\overline{\xi}_{ref} = [v_{x,ref}, x_{ref}, y_{ref}]^T$. $v_{x,ref}$ represents the reference longitudinal velocity along the reference trajectory. N_P is the prediction horizon. α_f and α_r represent the slip angles of the front and rear tires, respectively. The optimization objective function Eq. (16a) aims to minimize the cost from two aspects: (i) the trajectory-tracking error, and (ii) the control efforts, weighted by the positive-definite matrices $Q \in \mathbb{R}^{3\times3}$ and $R \in \mathbb{R}^{2\times2}$. Eqs. (16b)-(16c) impose constraints on the control inputs and the AV's handling stability [2].

Furthermore, the weight coefficients $Q = \text{diag}(Q_1, Q_2, Q_3)$ and $R = \text{diag}(R_1, R_2)$ would impact the tracking performance. In this work, we use a search method to determine the specific value [68]. By comparing test results with different weights, $Q_2 = Q_3 = 10Q_1 = 300$ and $R_1 = 5R_2 = 0.5$ are selected to offer a better trade-off among multi-objectives.

V. Simulation Experiments

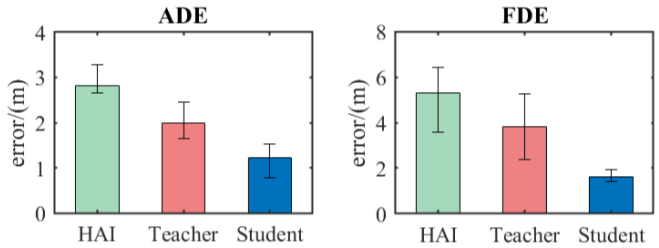
This section conducts simulation experiments to validate the proposed framework. The experiment results demonstrate the effectiveness of both the prediction module and the path planning and trajectory tracking module.

A. Scenarios and Datasets

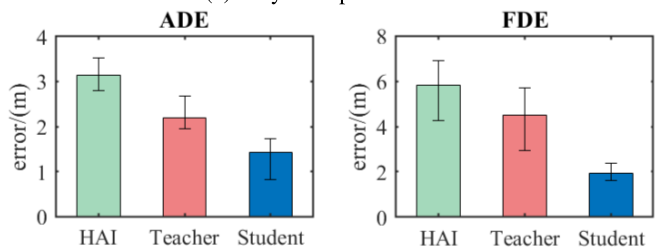
We perform the experiments in two scenarios: (i) the V2V interaction scenarios in the Waymo Open Dataset and (ii) a V2V interaction scenario in the driver simulator, where an AV operates along the planned path designed in Section IV, while a human driver performs the lane-change maneuver (see Fig. 11). For the driver simulator scenario, the commercial software CarSim is used to create a virtual simulation environment, where the HDV is controlled by a driver using an external steering wheel and accelerator pedal, and the AV is controlled with our proposed framework implemented using Matlab/Simulink.

The reason for considering these scenarios is two-fold. First, this shows that our proposed interaction-aware trajectory prediction algorithm can be used in different scenarios

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

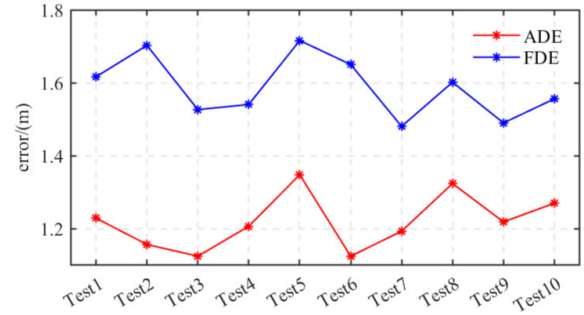


(a) Waymo Open Dataset.

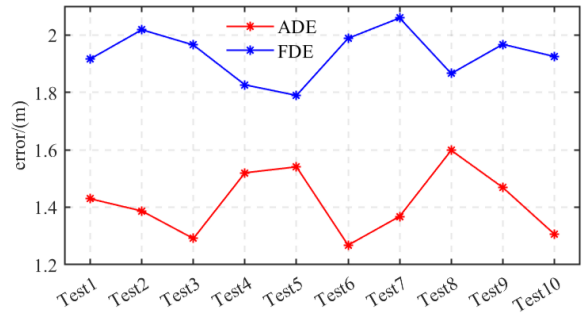


(b) Driver Simulator Dataset.

Fig. 12. Comparison of trajectory prediction accuracy with different networks.



(a) Waymo Open Dataset.



(b) Driver Simulator Dataset.

Fig. 13. Comparison of trajectory prediction accuracy across 10-fold cross validation.

with various human driving behavior and AV control algorithms. Second, the driver simulator experiment is used to validate the driving performance of integrating the proposed interaction-aware trajectory prediction algorithm and our proposed path-planning and motion control algorithms. Note that the interactions between HDVs and AVs rely on the control algorithms of AVs, i.e., the HDV behavior may change if the AV controller changes. Therefore, we cannot use the Waymo Open Dataset for evaluating the integration of our interaction-aware trajectory prediction and control algorithms, as Waymo data is generated by different and unknown AV control algorithms.

B. Performance of the Transformer-Transfer Learning-Based Predictor

In this subsection, we evaluate the performance of the transformer-transfer learning-based interaction-aware trajectory predictor. To assess the predictive accuracy, we employ two indicators for trajectory prediction: the minimum Average Distance Error (ADE) and the minimum Final Distance Error (FDE). The ADE calculates the average deviation of each sampling point in the predicted trajectories from the true reference, whereas the FDE quantifies the deviation between the final point of the predicted trajectories and the true reference. The prediction errors for two indicators in this work are calculated across all validation data.

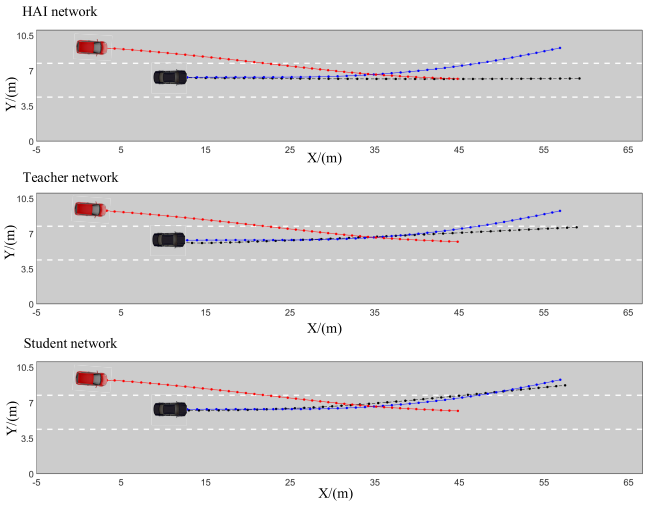
To facilitate the training of the proposed predictor on both datasets, we normalize the trajectory data by setting the longitudinal position of the AV as the origin and marking the lateral position based on the lanes. Then the trajectory data are divided into two parts: (i) separate HDVs' interaction scenarios involving lane-exchange maneuvers, and (ii) HDV-AV interaction scenarios involving lane-exchange maneuvers. The former data is used to train the trajectory predictor and serves as the teacher network. Then, the teacher network would be employed to guide the learning process of the student network using the latter data. For the training of the teacher network, we use 80% of the trajectory data to train the network, and the remaining data are used as the validation data. In terms of the training process of the student network, due to the limited autonomous driving scenarios, 90% of the trajectory data is employed to train the network, while the remaining data are used as validation data. Table II presents the average values of the performance indicators, as represented by Eqs. (13)-(15) in Section IV-B, for trajectory datasets (i) and (ii) selected from the Waymo dataset. The results indicate that, when interacting with AVs, HDVs prioritize safety and stability, often at the cost of driving comfort. In contrast, when two HDVs interact, driving comfort is prioritized, leading to a trade-off with safety and stability.

TABLE II
PERFORMANCE INDICATORS FOR DIFFERENT TRAJECTORY DATASETS

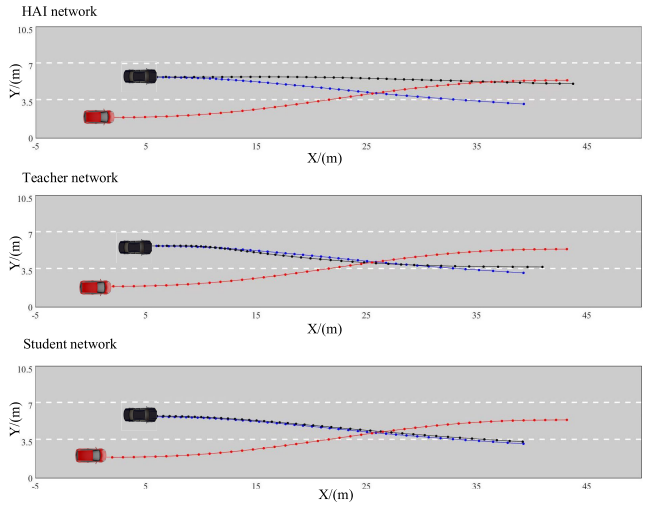
Performance indicators	Safety	Stability	Driving comfort
HDV-HDV interaction	9642.242	0.672	1329.865
AV-HDV interaction	8501.561	0.591	1478.417

1) Value of Using Transfer Learning Method: We show the value of using transfer learning method by comparing three benchmarks: (i) the trained teacher network (i.e., without transfer learning), (ii) the trained student network (i.e., the proposed training framework with transfer learning), and (iii) a trained network using only the trajectory data of the HDV-AV interaction scenarios (HAI network). The test results are shown in Fig. 12a and Fig. 12b for the Waymo dataset and the driver simulator dataset. From these results, it is clear that the student network (i.e., the proposed training framework) exhibits the highest predictive accuracy among the three methods. For the

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

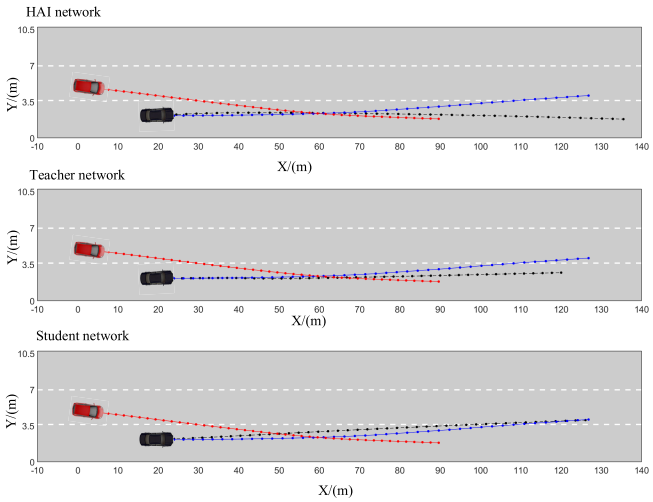


(a) Waymo Open Dataset.

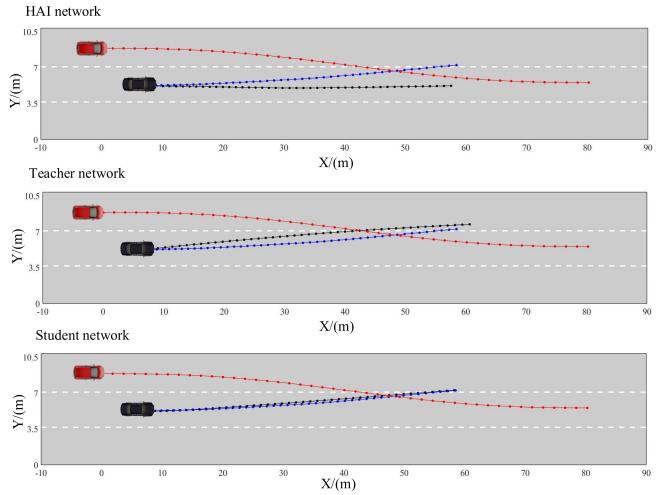


(b) Driver Simulator Dataset.

Fig. 14. The trajectory prediction results in a low-speed driving scenario.



(a) Waymo Open Dataset.



(b) Driver Simulator Dataset.

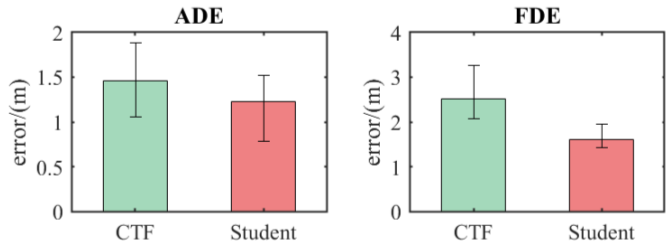
Fig. 15. The trajectory prediction results in a high-speed driving scenario.

Waymo dataset, the indicator of average ADE can be reduced by 56.26% and 37.42%, while the indicator of average FDE can be reduced by 69.60% and 56.86% compared to the HAI network and teacher network, respectively. For the driver simulator dataset, the average ADE shows reductions of 54.33% and 34.97%, while the average FDE demonstrates reductions of 70.02% and 57.45%, compared to the HAI network and teacher network, respectively. This demonstrates that the transformer-transfer learning-based framework effectively improves predictor accuracy in AV-HDV interaction scenarios despite limited training data. In addition, since we can collect more trajectory data from HDVs than AVs, the prediction results of the trained teacher network perform better than those of the HAI network.

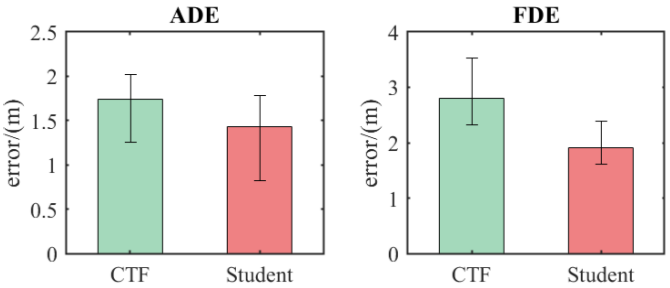
Furthermore, to mitigate the overfitting risk associated with the limited AV-HDV interaction data, we employ 10-fold cross-validation to evaluate the student network. This setup ensures that all samples are used for both training and testing across different folds, providing a more reliable and unbiased estimate of generalization performance. The average ADE and FDE across the 10 folds are presented in Fig. 13a and Fig. 13b, respectively. The results show that both ADE and FDE remain consistently low, with only minor variance across folds. This confirms that the model generalizes well to AV-HDV interaction scenarios and demonstrates the robustness and reliability of the proposed prediction network.

To visually present the prediction results of different networks, we perform two V2V driving scenarios on both datasets: (i) a high-speed driving scenario and (ii) a low-speed driving scenario. The related results are shown in Fig. 14a, Fig. 15a, Figs. 14b, and 15b respectively. The red line represents the AV's trajectories during the V2V interaction scenarios, while the blue line and the black line indicate the HDV's real trajectory and predicted trajectory, respectively. It is clear the proposed predictor can achieve better trajectory prediction results. Furthermore, we can observe that the prediction accuracy slightly decreases in the high-speed driving scenario compared to the low-speed driving scenario. This is because AVs mainly operate at lower speeds during

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

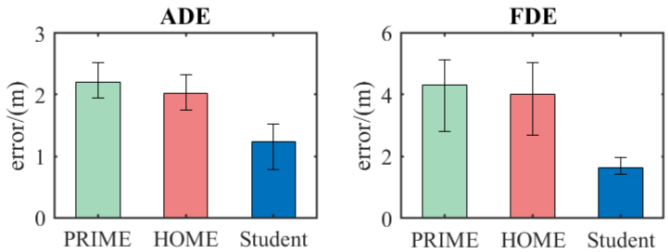


(a) Waymo Open Dataset.

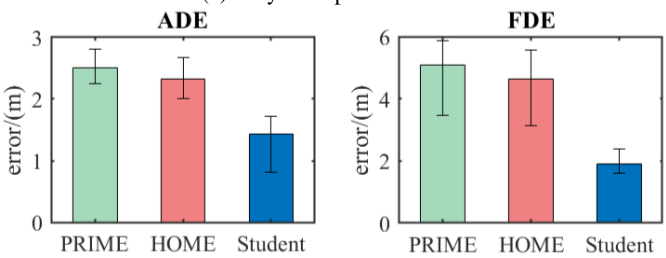


(b) Driver Simulator Dataset.

Fig. 16. Comparison of trajectory prediction accuracy with different transfer learning methods.



(a) Waymo Open Dataset.



(b) Driver Simulator Dataset.

Fig. 17. Comparison of trajectory prediction accuracy with different interactive trajectory predictors.

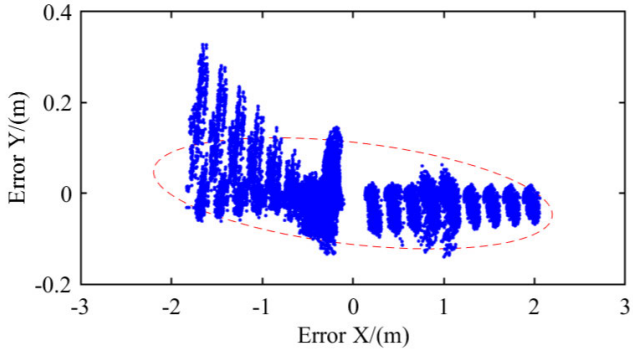


Fig. 18. The uncertain representation of the predictor on the simulator test data.

lane-change maneuvers, resulting in limited training data for high-speed driving scenarios.

2) Value of Using Hint-Based Transfer Learning Method: To assess the performance of the proposed transfer-learning framework during the training process, we compare prediction results between the proposed method (i.e., student network trained with a hint-based transfer learning method) and a classic transfer-learning training framework (CTF) that employs only the knowledge distillation method. As shown in Fig. 16a and Fig. 16b, it is clear that the proposed transfer learning method can significantly improve prediction accuracy by combining the hint-based training process and the classical knowledge distillation method. The indicators of ADE and FDE are reduced by 15.95% and 36.80%, respectively, for the Waymo dataset, and by 17.92% and 29.47%, respectively, for the driver simulator dataset.

3) Value of Using Different Interactive Trajectory Predictors: We further compare the performance of our proposed trajectory predictor with that of existing commonly used interactive trajectory predictors, including the PRIME [69] and HOME [70] algorithms. From the comparison results in Fig. 17a and 17b, it is clear that the values of ADE and FDE are relatively lower on the Waymo dataset and the driving simulator dataset when using the student network. This demonstrates that the proposed trajectory prediction framework can better ensure prediction accuracy.

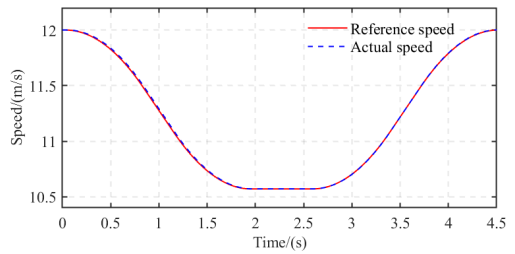
4) Runtime Analysis: Furthermore, we conduct a real-time performance analysis of the proposed trajectory predictor, which is executed on a Tesla A800 SXM GPU. Specifically, we use batch inference, where all planned trajectories are processed together in a batch, with repeated environmental context tensors aligned to each query. From the test results, its inference time is 160-175ms. Since behavior planning operates at a lower frequency ($\leq$2 Hz), the computation time meets real-time requirements.

C. Verification of Introducing Uncertain Prediction Results Into the Path Planning Process

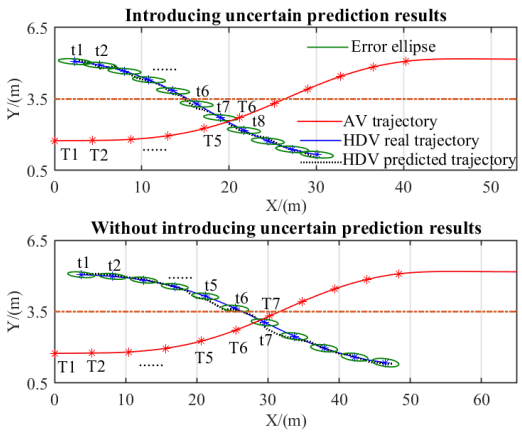
We verify the effectiveness of introducing uncertain prediction results, i.e., inequality 12, into the path planning process. Since the verification requires simulating the real-time interactions between the AV and HDV, we perform driver simulator-based experiments on V2V interaction scenarios with low-speed and high-speed driving conditions. We compare the motion control performance of the AV, including safety, stability, and comfort, with and without the incorporation of uncertainty quantification results into the path planning design. Here, we first use the driver simulator data to construct an error ellipse of the proposed trajectory predictor, as shown in Fig. 18.

1) Low-Speed Driving Scenario: The test results in the low-speed driving scenario are demonstrated in Fig. 19. It is clear from the speed profile shown in Fig. 19a that the proposed motion control can guarantee satisfactory longitudinal tracking performance. Fig. 19b illustrates the trajectories of the AV and HDV during a V2V interaction scenario involving lane-exchange maneuvers, including the predicted results from the trajectory predictor. The results show that the trajectory

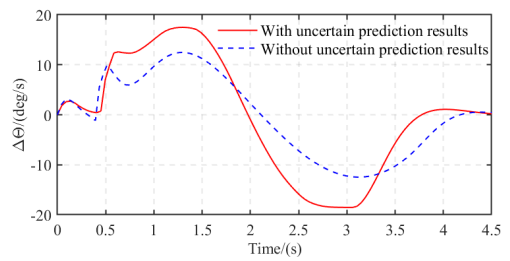
Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.



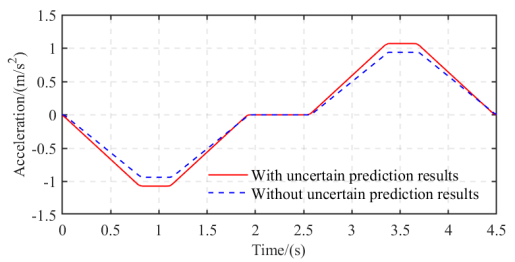
(a) The speed profile.



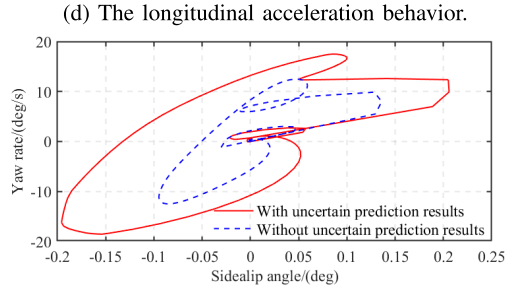
(b) The trajectories of the AV and the HDV.



(c) The change rate of the heading angle.



(d) The longitudinal acceleration behavior.



(e) The phase plane of AV's lateral motion.

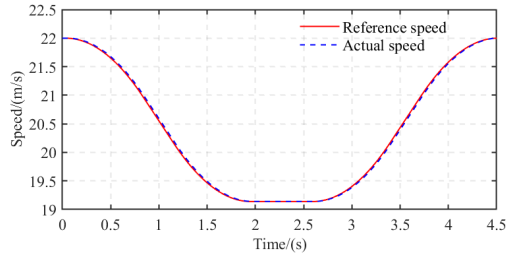
Fig. 19. Control performance in Test 1.

predictor's error consistently remains within the error ellipse shown in Fig. 18. This demonstrates the effectiveness of the proposed uncertainty quantification method in describing the predictor's errors. We label the error ellipse along the predicted trajectories in the tests, as depicted in Fig. 18. When the AV's trajectory passes through the error ellipse, it indicates a potential collision between vehicles. Furthermore, to better illustrate the results, we label the trajectory of the AV every three sampling instants as T_i, while the trajectory of the HDV every three sampling instants is labeled as t_i.

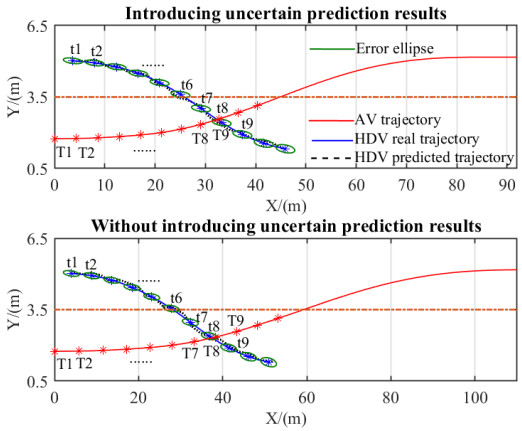
The path-planning ability with and without incorporating uncertain prediction results are compared. From the results, the AV can maintain a safe distance from the HDV when uncertain prediction results are introduced. For comparison, when uncertain prediction results are not introduced, the AV's trajectory approaches the error ellipse at the 7th sampling instant. However, as shown in inequality (12), in addition to the constraints of the error ellipse, a safety threshold S_{saf} must also be satisfied to guarantee a safe distance between vehicles. Hence, in this case, this trajectory cannot satisfy the safety constraint. Since uncertain prediction results are not introduced as a constraint, the TOPSIS strategy would select a trajectory with better performance indicators as proposed in section IV-C. This can be demonstrated in Figs. 19c-19e. In terms of driving comfort shown in Figs. 19c and 19d, due to a longer longitudinal lane-change distance as depicted in Fig. 19b, the AV exhibits a smaller change rate of heading angle and a smaller longitudinal deceleration behavior when uncertain prediction results are not considered, which represents better driving comfort. Furthermore, we introduce the classical "yaw rate-sideslip angle" phase plane [49] to describe the AV's stability performance. As illustrated in Fig. 19e, the AV operates within a smaller region in the phase plane when uncertain prediction results are not considered, indicating a more stable state. However, while relaxing the safety constraints of the AV may improve performance, it may also lead to potential collisions. This proves the effectiveness of introducing uncertain prediction results into the path-planning module to enhance the AV's safety.

2) High-Speed Driving Scenario: Another test case under a high-speed driving condition is also conducted. Fig. 20a demonstrates the satisfactory longitudinal-motion tracking performance. From the comparative tests of trajectory profiles in Fig. 20b, the error ellipse can accurately describe the uncertainty in prediction results. Furthermore, it is clear that the AV directly passes through the error ellipse at the 8th sampling instant when uncertain prediction results are not considered. This means that a potential collision could occur between vehicles when completing the lane-exchange maneuvers at high-speed driving conditions. The test results in Figs. 20c-20e also demonstrate that, when disregarding the constraint of vehicle safety distance, the proposed TOPSIS strategy would generate a reference path that is significantly more stable and comfortable. These findings are consistent with the test results in the low-speed driving scenario. The test results in both low-speed and high-speed driving scenarios demonstrate the effectiveness of the proposed path planning in ensuring vehicle safety

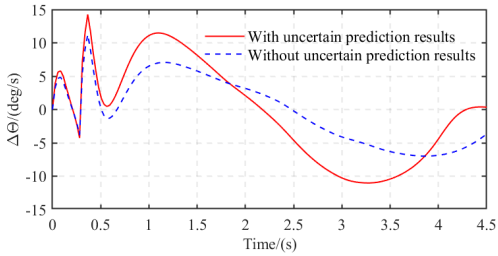
Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.



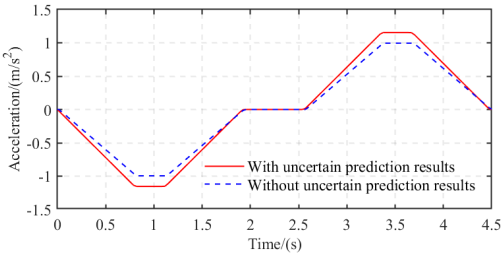
(a) The speed profile.



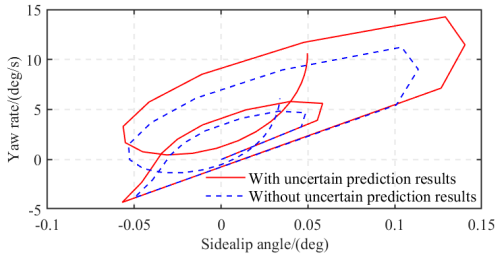
(b) The trajectories of the AV and the HDV.



(c) The change rate of the heading angle.



(d) The longitudinal acceleration behavior.



(e) The phase plane of AV's lateral motion.

Fig. 20. Control performance in test 2.

during V2V interaction scenarios involving lane-exchange maneuvers.

VI. Conclusion

In this work, we propose a systematic framework for motion planning in a V2V interaction scenario, informed by an interaction-aware trajectory prediction algorithm. We further improve the performance of the conventional interaction-aware predictor by proposing an innovative transform-transfer learning-based training method. Specifically, we construct a transfer learning framework to transfer knowledge from a teacher network (i.e., the conventional trajectory predictor) to the student framework, utilizing limited interaction data in autonomous driving scenarios. Subsequently, we quantify the uncertainties in the prediction results characterized as an error ellipse and introduced into the path-planning process to impose safety constraints when evaluating the performance of different candidate paths.

The test results demonstrate the proposed framework can be feasible for motion planning in a V2V interaction scenario involving lane-exchange maneuvers. Compared with conventional methods (i.e., the teacher network), the proposed trajectory predictor can improve the performance indicators of ADE and FDE by 34.97% and 57.45%, respectively, in the simulator test data. This study assumes perfect detection information and high-quality interaction data, which may limit its applicability in diverse scenarios. Future work will focus on improving model generalization. In addition, we plan to explore trajectory-level data augmentation techniques to further improve robustness, especially under limited-data conditions. Furthermore, considering the uncertainty of drivers' behaviors in multi-agent scenarios, future research can integrate more traffic information into the interaction-aware predictor to improve the performance, such as the driver's states.

Acknowledgment

This article solely reflects the opinions and conclusions of its authors and not Singapore MOE or any other entity.

References

[1] Đ. Petrović, R. Mijailović, and D. Pešić, "Traffic accidents with autonomous vehicles: Type of collisions, manoeuvres and errors of conventional vehicles' drivers," *Transp. Res. Proc.*, vol. 45, pp. 161–168, May 2020.

[2] J. Liang et al., "A MAS-based hierarchical architecture for the cooperation control of connected and automated vehicles," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 1559–1573, Feb. 2023.

[3] J. Liang, Q. Tian, J. Feng, D. Pi, and G. Yin, "A polytopic model-based robust predictive control scheme for path tracking of autonomous vehicles," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 2, pp. 3928–3939, Feb. 2024.

[4] Z. Xu, C. Wang, T. Jiang, Z. Liu, and N. Zheng, "Impediments to environmental awareness in autonomous driving systems and its effect on user adoption," *IEEE Trans. Intell. Vehicles*, early access, Mar. 8, 2024, doi: 10.1109/TIV.2024.3373773.

[5] W. Yue, C. Li, S. Wang, N. Xue, and J. Wu, "Cooperative incident management in mixed traffic of CAVs and human-driven vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 12462–12476, Jul. 2023.

[6] J. Zhou, L. Yan, and K. Yang, "Enhancing system-level safety in mixed-autonomy platoon via safe reinforcement learning," *IEEE Trans. Intell. Vehicles*, early access, Mar. 7, 2024, doi: 10.1109/TIV.2024.3373512.

[7] J. Zhou and K. Yang, "A parameter privacy-preserving strategy for mixed-autonomy platoon control," *Transp. Res. C, Emerg. Technol.*, vol. 169, Dec. 2024, Art. no. 104885.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

[8] M. Shawky, "Factors affecting lane change crashes," *IATSS Res.*, vol. 44, no. 2, pp. 155–161, Jul. 2020.

[9] H. Singh and A. Kathuria, "Analyzing driver behavior under naturalistic driving conditions: A review," *Accident Anal. Prevention*, vol. 150, Feb. 2021, Art. no. 105908.

[10] S. Mozaffari, O. Y. Al-Jarrah, M. Dianati, P. Jennings, and A. Mouzakitis, "Deep learning-based vehicle behavior prediction for autonomous driving applications: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 1, pp. 33–47, Jan. 2022.

[11] W. Xu, J. Wang, T. Fu, H. Gong, and A. Sobhani, "Aggressive driving behavior prediction considering driver's intention based on multivariate-temporal feature data," *Accident Anal. Prevention*, vol. 164, Jan. 2022, Art. no. 106477.

[12] X. Liao et al., "Driver digital twin for online prediction of personalized lane change behavior," *IEEE Internet Things J.*, vol. 10, no. 15, pp. 13235–13246, Aug. 2023.

[13] W. Wang, T. Qie, C. Yang, W. Liu, C. Xiang, and K. Huang, "An intelligent lane-changing behavior prediction and decision-making strategy for an autonomous vehicle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2927–2937, Mar. 2022.

[14] S. Lefèvre, D. Vasquez, and C. Laugier, "A survey on motion prediction and risk assessment for intelligent vehicles," *ROBOMECH J.*, vol. 1, no. 1, pp. 1–14, Dec. 2014.

[15] H. Gao et al., "Trajectory prediction of cyclist based on dynamic Bayesian network and long short-term memory model at unsignalized intersections," *Sci. China Inf. Sci.*, vol. 64, no. 7, Jul. 2021, Art. no. 172207.

[16] Q. Liu, S. Xu, C. Lu, H. Yao, and H. Chen, "Early recognition of driving intention for lane change based on recurrent hidden semi-Markov model," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 10545–10557, Oct. 2020.

[17] A. Girma, S. Amsalu, A. Workineh, M. Khan, and A. Homaifar, "Deep learning with attention mechanism for predicting driver intention at intersection," in *Proc. IEEE Intell. Vehicles Symp.*, Oct. 2020, pp. 1183–1188.

[18] D. J. Phillips, T. A. Wheeler, and M. J. Kochenderfer, "Generalizable intention prediction of human drivers at intersections," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2017, pp. 1665–1670.

[19] X. Chen, H. Zhang, Y. Hu, J. Liang, and H. Wang, "VNAGT: Variational non-autoregressive graph transformer network for multi-agent trajectory prediction," *IEEE Trans. Veh. Technol.*, vol. 72, no. 10, pp. 12540–12552, Oct. 2023.

[20] X. Chen, H. Zhang, F. Zhao, Y. Hu, C. Tan, and J. Yang, "Intention-aware vehicle trajectory prediction based on spatial–temporal dynamic attention network for Internet of Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19471–19483, Oct. 2022.

[21] B. Yang, F. Fan, R. Ni, H. Wang, A. Jafaripournimchahi, and H. Hu, "A multi-task learning network with a collision-aware graph transformer for traffic-agents trajectory prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 6677–6690, Jul. 2024.

[22] M. McNaughton, C. Urmson, J. M. Dolan, and J.-W. Lee, "Motion planning for autonomous driving with a conformal spatiotemporal lattice," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2011, pp. 4889–4895.

[23] W. Schwarting, J. Alonso-Mora, and D. Rus, "Planning and decision-making for autonomous vehicles," *Annu. Rev. Control*, vol. 1, no. 1, pp. 187–210, Jan. 2018.

[24] M. Wang, Z. Wang, J. Talbot, J. C. Gerdes, and M. Schwager, "Game-theoretic planning for self-driving cars in multivehicle competitive scenarios," *IEEE Trans. Robot.*, vol. 37, no. 4, pp. 1313–1325, Aug. 2021.

[25] C. Li, T. Trinh, L. Wang, C. Liu, M. Tomizuka, and W. Zhan, "Efficient game-theoretic planning with prediction heuristic for socially-compliant autonomous driving," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 10248–10255, Oct. 2022.

[26] W. Duan, Z. Tang, W. Liu, and H. Zhou, "Autonomous driving planning and decision making based on game theory and reinforcement learning," *Expert Syst.*, vol. 40, no. 5, p. 13191, Jun. 2023.

[27] K. Kurzer, C. Zhou, and J. M. Zöllner, "Decentralized cooperative planning for automated vehicles with hierarchical Monte Carlo tree search," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2018, pp. 529–536.

[28] J. Dinneweth, A. Boubezoul, R. Mandiau, and S. Espié, "Multi-agent reinforcement learning for autonomous vehicles: A survey," *Auto. Intell. Syst.*, vol. 2, no. 1, p. 27, Nov. 2022.

[29] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 961–971.

[30] N. Deo and M. M. Trivedi, "Convolutional social pooling for vehicle trajectory prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 1468–1476.

[31] H. Song, W. Ding, Y. Chen, S. Shen, M. Y. Wang, and Q. Chen, "PiP: Planning-informed trajectory prediction for autonomous driving," in *Proc. Eur. Conf. Comput. Vis.*, Glasgow, U.K., 2020, pp. 598–614.

[32] Y. Chen et al., "Efficient speed planning for autonomous driving in dynamic environment with interaction point model," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 11839–11846, Oct. 2022.

[33] J. L. V. Espinoza, A. Liniger, W. Schwarting, D. Rus, and L. Van Gool, "Deep interactive motion prediction and planning: Playing games with motion prediction models," in *Proc. Learn. Dyn. Control Conf.*, 2022, pp. 1006–1019.

[34] J. Ngiam et al., "Scene transformer: A unified architecture for predicting multiple agent trajectories," 2021, *arXiv:2106.08417*.

[35] K. Zhang, X. Feng, L. Wu, and Z. He, "Trajectory prediction for autonomous driving using spatial–temporal graph attention transformer," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22343–22353, Nov. 2022.

[36] A. Sadeghian, V. Kosaraju, A. Sadeghian, N. Hirose, H. Rezatofighi, and S. Savarese, "SoPhie: An attentive GAN for predicting paths compliant to social and physical constraints," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1349–1358.

[37] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese, and A. Alahi, "Social GAN: Socially acceptable trajectories with generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2255–2264.

[38] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[39] N. Lee, W. Choi, P. Vernaza, C. B. Choy, P. H. S. Torr, and M. Chandraker, "DESIRE: Distant future prediction in dynamic scenes with interacting agents," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 336–345.

[40] Y. Rahmati, M. K. Hosseini, A. Talebpour, B. Swain, and C. Nelson, "Influence of autonomous vehicles on car-following behavior of human drivers," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2673, no. 12, pp. 367–379, Dec. 2019.

[41] M. F. Ozkan and Y. Ma, "Modeling driver behavior in car-following interactions with automated and human-driven vehicles and energy efficiency evaluation," *IEEE Access*, vol. 9, pp. 64696–64707, 2021.

[42] R. Zhang, S. Masoud, and N. Masoud, "Impact of autonomous vehicles on the car-following behavior of human drivers," *J. Transp. Eng., A, Syst.*, vol. 149, no. 3, Mar. 2023, Art. no. 04022152.

[43] X. Wen, Z. Cui, and S. Jian, "Characterizing car-following behaviors of human drivers when following automated vehicles using the real-world dataset," *Accident Anal. Prevention*, vol. 172, Jul. 2022, Art. no. 106689.

[44] X. Li, Z. You, X. Ma, X. Pang, X. Min, and H. Cui, "Effect of autonomous vehicles on car-following behavior of human drivers: Analysis based on structural equation models," *Phys. A, Stat. Mech. Appl.*, vol. 633, Jan. 2024, Art. no. 129360.

[45] H. Xu, W. Xiao, C. G. Cassandras, Y. Zhang, and L. Li, "A general framework for decentralized safe optimal control of connected and automated vehicles in multi-lane signal-free intersections," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 17382–17396, Oct. 2022.

[46] M. Chen et al., "FaSTrack: A modular framework for real-time motion planning and guaranteed safe tracking," *IEEE Trans. Autom. Control*, vol. 66, no. 12, pp. 5861–5876, Dec. 2021.

[47] L. Zhang, R. Zhang, T. Wu, R. Weng, M. Han, and Y. Zhao, "Safe reinforcement learning with stability guarantee for motion planning of autonomous vehicles," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 12, pp. 5435–5444, Dec. 2021.

[48] G. Velasco-Hernandez, D. J. Yeong, J. Barry, and J. Walsh, "Autonomous driving architectures, perception and data fusion: A review," in *Proc. IEEE 16th Int. Conf. Intell. Comput. Commun. Process. (ICCP)*, Sep. 2020, pp. 315–321.

[49] J. Liang et al., "A distributed integrated control architecture of AFS and DYC based on MAS for distributed drive electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5565–5577, Jun. 2021.

[50] Z. Wang, J. Guo, Z. Hu, H. Zhang, J. Zhang, and J. Pu, "Lane transformer: A high-efficiency trajectory prediction model," *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 2–13, 2023.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.

[51] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.
[52] A. Veit, M. J. Wilber, and S. Belongie, "Residual networks behave like ensembles of relatively shallow networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, Dec. 2016, pp. 550–558.
[53] Y. Huang, J. Du, Z. Yang, Z. Zhou, L. Zhang, and H. Chen, "A survey on trajectory-prediction methods for autonomous driving," *IEEE Trans. Intell. Vehicles*, vol. 7, no. 3, pp. 652–674, Sep. 2022.
[54] J.-H. Bae, J. Yim, N.-S. Kim, C.-S. Pyo, and J. Kim, "Layer-wise hint-based training for knowledge transfer in a teacher–student framework," *ETRI J.*, vol. 41, no. 2, pp. 242–253, Apr. 2019.
[55] R. Adriana, B. Nicolas, K. S. Ebrahimi, C. Antoine, G. Carlo, and B. Yoshua, "Fitnets: Hints for thin deep nets," in *Proc. ICLR*, 2015, vol. 2, no. 3, pp. 1–10.
[56] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
[57] G. Chen, W. Choi, Y. Xiang, T. X. Han, and M. Chandraker, "Learning efficient object detection models with knowledge distillation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Dec. 2017, pp. 742–751.
[58] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
[59] M. Cao, R. Wang, J. Wang, and N. Chen, "An integrated MPC approach for FWIA autonomous ground vehicles with emergency collision avoidance," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2018, pp. 2432–2437.
[60] H. O. Lancaster and E. Seneta, "Chi-square distribution," in *Encyclopedia Biostatistics*, vol. 2. New York, NY, USA: Wiley, 2005.
[61] C. Huang et al., "Personalized trajectory planning and control of lane-change maneuvers for autonomous driving," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5511–5523, Jun. 2021.
[62] Y. Li, L. Li, and D. Ni, "Dynamic trajectory planning for automated lane changing using the quintic polynomial curve," *J. Adv. Transp.*, vol. 2023, pp. 1–14, Sep. 2023.
[63] J. R. G. Martínez, J. R. Reséndiz, M. Á. M. Prado, and E. E. C. Miguel, "Assessment of jerk performance S-curve and trapezoidal velocity profiles," in *Proc. 8th Int. Eng. Congr. (CONIIN)*, May 2017, pp. 1–7.
[64] J. Nilsson, M. Brännström, E. Coelingh, and J. Fredriksson, "Lane change maneuvers for automated vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 5, pp. 1087–1096, May 2017.
[65] H. Vijayakumar et al., "A holistic safe planner for automated driving considering interaction with human drivers," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 1, pp. 2061–2076, Jan. 2024.
[66] Y. Ding, W. Zhuang, L. Wang, J. Liu, L. Guvenc, and Z. Li, "Safe and optimal lane-change path planning for automated driving," *Proc. Inst. Mech. Eng., D, J. Automobile Eng.*, vol. 235, no. 4, pp. 1070–1083, Mar. 2021.
[67] Z. Li, Z. Luo, Y. Wang, G. Fan, and J. Zhang, "Suitability evaluation system for the shallow geothermal energy implementation in region by entropy weight method and TOPSIS method," *Renew. Energy*, vol. 184, pp. 564–576, Jan. 2022.
[68] J. Liang et al., "An energy-oriented torque-vector control framework for distributed drive electric vehicles," *IEEE Trans. Transport. Electrific.*, vol. 9, no. 3, pp. 4014–4031, Mar. 2023.
[69] H. Song, D. Luan, W. Ding, M. Y. Wang, and Q. Chen, "Learning to predict vehicle trajectories with model-based planning," in *Proc. Conf. Robot Learn.*, 2022, pp. 1035–1045.
[70] T. Gilles, S. Sabatini, D. Tsishkou, B. Stanciulescu, and F. Moutarde, "HOME: Heatmap output for future motion estimation," in *Proc. IEEE Int. Intell. Transp. Syst. Conf. (ITSC)*, Sep. 2021, pp. 500–507.

Jinhao Liang (Member, IEEE) received the B.S. degree from the School of Mechanical Engineering, Nanjing University of Science and Technology, Nanjing, China, in 2017, and the Ph.D. degree from the School of Mechanical Engineering, Southeast University, Nanjing, in 2022. He is currently a Research Fellow with the Department of Civil and Environmental Engineering, National University of Singapore. His research interests include vehicle dynamics and control, autonomous vehicles, and vehicle safety assistance systems.

Chaopeng Tan received the B.S. and Ph.D. degrees in traffic engineering from Tongji University, Shanghai, China, in 2017 and 2022, respectively. He was a Post-Doctoral Research Fellow with the Department of Civil and Environmental Engineering, National University of Singapore, from 2022 to 2024, and a Post-Doctoral Researcher with the Department of Transport and Planning, Delft University of Technology, The Netherlands. He is currently a Post-Doctoral Research Associate with the Chair of Traffic Process Automation, Technische Universität Dresden, Germany. His main research interests include intelligent transportation systems, traffic modeling and control with connected vehicles, privacy-preserving traffic control, and multi-modal traffic management.

Longhao Yan (Member, IEEE) received the B.Eng. and M.Eng. degrees from the School of Electronics and Control Engineering, Chang'an University, Xi'an, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree with the National University of Singapore. His research interests include lateral control and trajectory prediction of intelligent transportation systems.

Jingyuan Zhou (Graduate Student Member, IEEE) received the B.Eng. degree in electronic information science and technology from Sun Yat-sen University, Guangzhou, China, in 2022. He is currently pursuing the Ph.D. degree with the National University of Singapore. His research interests include safe and secure connected and automated vehicles control in intelligent transportation systems.

Guodong Yin (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from Southeast University, Nanjing, China, in 2007. From August 2011 to August 2012, he was a Visiting Scholar with the Department of Mechanical and Aerospace Engineering, The Ohio State University, Columbus, OH, USA. He is currently a Professor with the School of Mechanical Engineering, Southeast University. His research interests include vehicle dynamics and control, automated vehicles, and connected vehicles. He was a recipient of the National Science Fund for Distinguished Young Scholars.

Kaidi Yang (Member, IEEE) received the B.Eng. and M.Sc. degrees from Tsinghua University and the Ph.D. degree from ETH Zurich. He is an Assistant Professor with the Department of Civil and Environmental Engineering, National University of Singapore. Prior to this, he was a Post-Doctoral Researcher with the Autonomous Systems Laboratory, Stanford University. His main research interest is the operation of future mobility systems enabled by connected and automated vehicles (CAVs) and shared mobility.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 10:52:20 UTC from IEEE Xplore. Restrictions apply.